

Reference model-based optimization of sliding mode control for second-order systems

Measurement and Control

2026, Vol. 59(6) 795–812

© The Author(s) 2025

Article reuse guidelines:

sagepub.com/journals-permissions

DOI: 10.1177/00202940251355057

journals.sagepub.com/home/mac

İbrahim Halil Teke^{1,2} and Nusret Tan¹

Abstract

In this study, the design of Sliding Mode Control (SMC) has been carried out based on a second order reference model with a specified transfer function. The primary objective of this study is to determine the SMC controller parameters according to the desired time response in the output signal. The aim is to determine the optimal control gains that minimize the difference between the reference and plant outputs in a systematic way. By minimizing the error between the reference model and the SMC controlled system model, the optimal SMC parameters, specifically the sliding surface slope (λ) and the switching function coefficient (K), are obtained. Integral performance criteria were used to minimize the error. Both the reference system and the system controlled by the SMC have second-order transfer functions. While the natural frequency (ω_n) in the reference model system remains constant, the damping ratio (ζ) was varied between 0.5 and 1. For each ζ value in the reference system, the optimal SMC parameters (λ and K) were obtained by using the Fmincon optimization algorithm to minimize the error between the two systems based on the selected integral performance criteria. By applying the obtained SMC parameters to the controlled system, unit step responses were obtained in the output of the controlled system. The optimization process was performed based on the transfer functions of two different known systems, and the variation of the optimal SMC parameters according to the damping ratio of the reference model was presented in tables and graphs. The simulation results confirm that the proposed method significantly improves the accuracy and robustness of tracking compared to conventional SMC tuning approaches. These results demonstrate the effectiveness of the proposed method for designing reliable and efficient controllers for second-order dynamic systems. The success of the reference model-based second-order system controlled by the SMC method is clearly demonstrated by the generated graphs and tables. This study introduces a new optimization-based SMC design approach by establishing an explicit analytical relationship between the damping ratio and the controller parameters. In the proposed method uses regression, which avoids the need for iterative tuning and repeated calibration.

Keywords

sliding surface, control rule, integral performance criteria, fmincon, curve fitting

Date received: 24 February 2025; accepted: 26 May 2025

Introduction

In modern control systems, various control methods have been developed to optimize system performance and effectively manage dynamic behaviors. Traditional control methods, such as PI, PD, and PID controllers, provide simple and widely used solutions in practical applications. However, they exhibit significant sensitivity to nonlinear dynamics and uncertainties in system parameters, which is a critical disadvantage. This situation has highlighted the necessity of developing more robust and flexible control methods. In this context, SMC has emerged as a prominent approach.

SMC significantly enhances control performance in both linear and nonlinear systems by providing robustness against disturbances and uncertainties. However, a

major drawback of this method is the “chattering” problem, which can adversely affect system stability, making its resolution crucial. Moreover, optimizing SMC parameters has the potential to further improve control performance. The effectiveness of SMC in different applications has been extensively analyzed in the literature. Although traditional control methods such

¹Electrical and Electronics Engineering Department, Inonu University, Malatya, Turkey

²Afşin Elbistan B Thermal Power Plant, EUAŞ, Kahramanmaraş, Turkey

Corresponding author:

İbrahim Halil Teke, Electrical and Electronics Engineering Department, Inonu University, Malatya 44280, Turkey; Afşin Elbistan B Thermal Power Plant, EUAŞ, Kahramanmaraş, Turkey.

Email: ibrahim.halil.teke@gmail.com



Creative Commons CC BY: This article is distributed under the terms of the Creative Commons Attribution 4.0 License (<https://creativecommons.org/licenses/by/4.0/>) which permits any use, reproduction and distribution of the work without

further permission provided the original work is attributed as specified on the SAGE and Open Access pages (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

as PI, PD, and PID controllers are widely used to meet common requirements, they fail to provide fast and stable output voltage under non-nominal variable loads. In such cases, SMC offers an alternative solution. Developed in the 1970s, SMC is frequently applied in fields such as robotics, power electronics, and servo drive control.

SMC provides robust control for both linear and nonlinear systems by showing insensitivity to parameter variations.¹ In a study comparing PID and SMC controllers, SMC demonstrated superior performance in terms of settling time, peak overshoot, and robustness.² For an inverted pendulum system, SMC improved the balancing time by 54.3%, showing better performance than PID in nonlinear systems.³ Terminal sliding mode controllers ensure that the error converges to zero in a finite time and enhance system stability.⁴ Additionally, PID-SMC and SMC methods with sigmoid functions have significantly reduced the chattering problem, leading to superior performance.⁵ In DC motor speed control under loaded conditions, SMC demonstrated higher robustness against parameter variations and outperformed PID controllers.⁶ In photovoltaic pump systems, SMC and field-oriented control methods effectively reduced chattering problems.⁷ For permanent magnet synchronous motors (PMSM), the SMC + PID strategy achieved better results compared to classical SMC.⁸ In wind turbine emulators, SMC outperformed PI controllers in reference current tracking and reducing settling time.⁹ Fast terminal sliding mode control provided an effective solution under uncertainties and disturbances, ensuring the system reached equilibrium in a shorter time.¹⁰ Fuzzy logic-based SMC has been proven to offer efficient solutions in engineering problems.¹¹ Studies on the practical applicability of conventional SMC techniques have shown that the chattering effect can be minimized with proper parameter tuning.¹² SMC has been found effective in robotic manipulators and nonlinear systems in terms of fast convergence and disturbance rejection.¹³ Data-driven SMC methods have shown higher robustness compared to traditional methods.¹⁴ Model-Free Sliding Mode Control (MFSMC) approaches offer efficient solutions for nonlinear systems.¹⁵

In continuous-time nonlinear systems, Continuous Model-Free SMC (CMFSMC) has provided control by reducing the chattering effect.¹⁶ In electromechanical actuators, Model Reference Adaptive SMC (MRDASMC) has demonstrated high performance and robustness against parameter changes and external disturbances.¹⁷ Dynamic SMC and adaptive PID controllers have addressed the shortcomings of SMC and improved its performance.¹⁸ The Second Order Sliding Mode Controller (SOSMC) has provided solutions to load frequency control problems by minimizing the chattering effect.¹⁹ To reduce the chattering effect, Adaptive Neuro-Fuzzy Inference Systems (ANFIS) combined with SMC have achieved high-accuracy control.²⁰ Furthermore, sliding mode observers have been

more effective in suppressing disturbances in control structures.²¹ Hybrid SMC approaches have proven to be effective methods in achieving energy-efficient control and high robustness in nonlinear systems.²² In smart power systems, supporting SMC with artificial neural networks has enhanced learning capability and accelerated the error correction process.²³ The durability-enhancing effect of SMC in semiconductor-based power electronics circuits has been investigated, yielding positive results.²⁴ Moreover, sliding mode-based learning algorithms in electromechanical systems have provided adaptive control and improved performance.²⁵ In conclusion, SMC methods offer a superior alternative to PID controllers in managing uncertainties and providing robustness in nonlinear systems. SMC stands out as an effective control strategy in engineering applications, both in theory and practice.²⁶ Furthermore, recent studies emphasize the increasing importance of intelligent control techniques in improving system performance. For example, Puentes et al.²⁷ proposed neural network-based intelligent controllers that significantly improve trajectory tracking performance in mobile robotics. Similarly, Baghdasaryan and Hovhannisyan introduced a machine learning-based stability assessment model for electric drive systems, demonstrating its effectiveness in fault diagnosis and robust control under uncertain operating conditions.²⁸ Additionally, Abidova²⁹ developed a new visualization and analysis method for defect detection in electromechanical equipment, utilizing singular spectral analysis to detect hidden defects even in the presence of heavy noise. Together, these studies collectively highlight the critical role of intelligent systems and machine learning techniques in advancing modern control system design. Recent advancements in sliding mode control (SMC) have introduced innovative solutions to improve control robustness and mitigate the effects of chattering in various dynamic systems. For example Hou et al.³⁰ proposed a finite-time continuous terminal sliding mode control (FT-CTSMC) strategy for servo motor systems. This strategy ensures finite-time stability independent of initial conditions and successfully reduces chattering through a continuous control approach. Rsetam et al.³¹ developed a GPIO-based continuous sliding mode control (CSMC) framework to address unmatched disturbances and network-induced time delays in networked control systems. The superior tracking performance of this framework was validated experimentally on servo motor platforms. In the context of flexible joint robotics, Khan et al.³² introduced a singular perturbation-based adaptive integral sliding mode control (SP-AISMC) method to achieve high-precision trajectory tracking despite parametric uncertainties and external disturbances. Zheng et al.³³ further advanced the field by integrating machine learning techniques and proposing an extreme learning machine-based super-twisting integral terminal sliding mode control (ST-ITSMC) method for permanent magnet synchronous motors. This method significantly

reduces chattering and guarantees finite-time convergence. Additionally, Rsetam et al.³⁴ designed a robust terminal sliding mode control (TSMC) approach for underactuated flexible joint robots, achieving finite-time convergence and strong robustness against nonlinearities and external disturbances. They also proposed a cascaded extended state observer (CESO)-based SMC method, which improves the accuracy of state estimation and capabilities of disturbance rejection while effectively suppressing noise amplification in high-order control systems.³⁵ Together, these recent studies demonstrate the importance of advanced SMC techniques and intelligent control integrations in achieving robust, high-performance and robust control solutions under varying system uncertainties. Although many of these studies demonstrate the advantages of SMC and various enhancements for improving performance, they often rely on heuristic tuning strategies or simulation-based parameter adjustments, which may not be easily applicable to different system configurations. In particular, there is a lack of a systematic framework in the literature that directly links reference model Dynamics, such as variations in the damping ratio, with optimal SMC parameters through formal optimization criteria. This limitation reveals a fundamental research gap that this study aims to address.

In this study, a reference system with a second-order transfer function and a system controlled by SMC were examined. The error performance between the systems outputs was optimized to determine the most suitable SMC parameters, specifically the sliding surface slope (λ) and the switching function coefficient (K). Integral performance criteria, including ISE, IAE, ITSE, and ITAE, were used for this purpose. The key contribution of this study is establishing a graphical and analytical relationship between the ζ and SMC parameters, thereby eliminating the need for infinite simulations. The study was conducted using the Fmincon optimization algorithm and Integral performance criteria method, followed by the Curve Fitting Method to derive linear polynomial functions relating with ζ to λ and ζ to K . These analyses were performed within the MATLAB platform to examine how SMC control parameters vary as the damping ratio changes between 0.5 and 1. System performance was evaluated across a broad spectrum, ranging from underdamped systems to critically damped systems. The uniqueness of this study is that it provides a new approach to optimizing SMC parameters by establishing a direct correlation with a second-order reference model. Unlike conventional tuning methods that rely on iterative simulations, our methodology leverages integral performance criteria and optimization techniques to systematically determine the ideal control parameters. By varying the damping ratio and analyzing its impact on system dynamics, this research provides a structured framework for designing high-performance controllers with minimal computational effort. The findings not only enhance the robustness and adaptability of SMC but

also pave the way for future advancements in intelligent control strategies for complex nonlinear systems.

This study's new lies in establishing an analytical correlation between the damping characteristics of the reference model and the optimized SMC parameters (λ and K) through a hybrid approach combining constrained optimization and regression modeling. Unlike adaptive reaching laws or conventional SMC designs, which rely on trial-and-error or online adaptation, the proposed framework provides an offline-tuned, equation-driven SMC design strategy that efficiently minimizes error metrics and suppresses chattering.

This contribution is motivated by the need for a systematic, model-based design of sliding mode controllers (SMCs) that goes beyond traditional heuristic tuning methods. Although SMCs are known for their robustness and wide applicability, their practical implementation often suffers from a lack of structured approaches to tuning parameters. In particular, the absence of an analytical link between the desired system dynamics (e.g. damping ratio) and the SMC parameters limits the design efficiency and scalability in real-world applications. This work introduces a new optimization framework that explicitly correlates the reference model's damping ratio with optimal SMC parameters using constrained optimization and curve-fitting techniques. The contribution of the paper is outlined below:

- An optimization-based SMC design methodology is introduced, that establishes a direct analytical relationship between the damping ratio and the controller parameters (λ and K). This eliminates the need for repetitive simulations.
- The proposed approach systematically determines optimal SMC parameters by minimizing four integral performance criteria (ISE, IAE, ITSE, and ITAE) using the MATLAB-based Fmincon optimization algorithm.
- Polynomial regression models are developed using the Curve Fitting Toolbox to efficiently predict controller parameters, enabling quick estimation of parameters for varying damping ratios without the need for additional simulations.
- The proposed method effectively reduces the chattering and improves tracking accuracy, as validated through extensive MATLAB/Simulink simulations.
- The framework is structured and scalable, and can be extended to higher-order and nonlinear systems, offering practical applicability for real-time control implementations.

Materials and methods

This section describes the methodology that integrates optimization, performance evaluation, and modeling to improve the effectiveness of the control system. The

study was carefully designed to explore the impact of damping ratio (ζ) of a second-order reference model on the optimal tuning of Sliding Mode Control (SMC) parameters. Specifically, the impact on the sliding surface slope (λ) and switching gain coefficient (K) was assessed using a reference system with a fixed natural frequency ($\omega_n = 1$ rad/s) and varying damping ratio values ranging from 0.5 to 1. To evaluate the system performance, integral performance criteria (ISE, IAE, ITSE, ITAE) were applied, allowing a quantitative comparison of different controllers in terms of accuracy and stability. The MATLAB-based `fmincon` optimization algorithm was employed to minimize each performance criterion for each damping ratio value, and the resulting optimal (λ) and (K) values were recorded for further analysis. In addition, the curve fitting method was used to model the system behavior, providing a better understanding of the relationships between the control variables. To establish a generalized and analytical link, polynomial regression models were constructed using the Curve Fitting Toolbox in MATLAB, mapping λ and K as functions of ζ . These models enabled predictive estimation without repeating simulation-based optimization. This approach ensures a robust and evaluation of the proposed control strategy.

Fmincon

In this study, the optimization archive function “`fmincon`” from the MATLAB program was used (MATLAB, The MathWorks). `Fmincon` is a function used in the MATLAB to solve constrained optimization problems. The proposed optimization framework is applied to a second-order system with a reference model having a constant natural frequency of 1 rad/s and a variable damping ratio.

This function minimizes a nonlinear function with multiple variables. `Fmincon` can include both linear and nonlinear constraints. It is employed to find the values of “ x ” that minimize a nonlinear multivariable function $f(x)$ under given constraints. In optimal design, the use of the “`fmincon`” archive function solves optimization problems with nonlinear objective functions under linear or nonlinear constraints by utilizing mathematical programming methods provided by the MATLAB program. In this study the function minimizes the error between the reference model and the SMC-controlled system while considering performance constraints. The MATLAB code snippet illustrating the implementation of constraints and initial conditions are;

```
% Fmincon Optimization Setup
lb = [0, 0]; % Lower bounds for  $\lambda$  and  $K$ 
ub = [100, 100]; % Upper bounds for  $\lambda$  and  $K$ 
x0 = [1, 1]; % Initial guess for  $\lambda$  and  $K$ 
options = optimoptions('fmincon', 'Display', 'iter', 'Algorithm', 'sqp');
```

```
[optParams, fval] = fmincon(@costFunction, x0, [], [], [], lb, ub, [], options);
```

Optimization algorithm and flowchart. The flowchart and algorithm presented below summarize the steps taken to optimize the SMC controller parameters (λ and K) using MATLAB’s `fmincon` algorithm. This structured procedure ensures the accurate minimization of error between the SMC-controlled system and the reference model.

Algorithm steps for Fmincon-based SMC optimization:

1. Define the second-order reference model in simulink with fixed $\omega_n = 1$ rad/s and varying damping ratio $\zeta \in [0.5, 1]$.
2. Set up the controlled system with SMC structure (equations (13) and (16)).
3. Initialize the λ ($k(1)$) and K ($k(2)$)
4. Define lower and upper bounds values for λ and K .
5. Define optimization options
6. Define the cost function using one of the performance criteria (ISE, IAE, ITSE, ITAE).
7. Use `fmincon` to minimize the error between the reference and controlled system outputs.
8. Store optimal λ and K , and the performance criteria values for each ζ .
9. Apply the Curve Fitter Toolbox to model λ (with ζ) and K (with ζ) using linear polynomial regression.

Figure 1 visually illustrates the optimization procedure described above, detailing the iterative process used to effectively determine the optimal controller parameters.

Integral performance criteria

In control systems, several specific performance criteria are used to measure and compare system performance. In this study, the evaluations were conducted based on four different criteria: Integral Square Error (ISE), Integral Absolute Error (IAE), Integral Time-weighted Square Error (ITSE), and Integral Time-weighted Absolute Error (ITAE). These performance criteria play a significant role in comparing controllers in terms of accuracy, response time, and stability. The calculation formulas for these criteria are expressed in equations (1)–(4) as follows.^{2,23}

$$ISE = \int_0^T e(t)^2 dt \quad (1)$$

$$IAE = \int_0^T |e(t)| dt \quad (2)$$

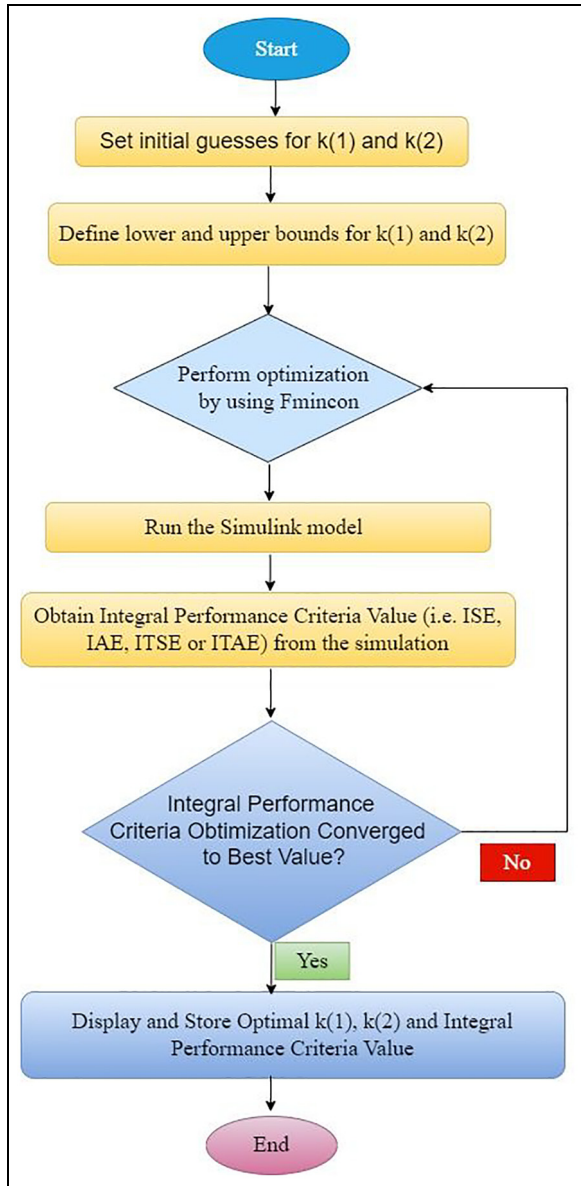


Figure 1. Flowchart of the proposed Fmincon-based SMC optimization framework.

$$ITSE = \int_0^T t \cdot e(t)^2 dt \quad (3)$$

$$ITAE = \int_0^T t \cdot |e(t)| dt \quad (4)$$

The selection of ISE, IAE, ITSE, and ITAE as performance criteria in this study was motivated by their complementary characteristics in evaluating system performance. These metrics are commonly employed in control theory to capture various aspects of error dynamics, including amplitude, duration, and timing. Specifically, ISE focuses on minimizing large deviations; IAE provides a balanced measure of total accumulated error; ITSE targets rapid transient behavior; and ITAE prioritizes minimization of prolonged errors.

Collectively, these criteria facilitate a comprehensive assessment of performance, making them particularly suitable for fine-tuning controller parameters in second-order systems. ISE involves the integration of the square of the error signal over time. A smaller ISE value indicates that the system is capable of reducing large errors more quickly. This criterion focuses on minimizing significant deviations from the desired output and ensures that the system performs well in terms of error correction and stability. IAE represents the integration of the absolute value of the error signal over time, a lower IAE value indicates that the overall amount of error in the system is smaller. This criterion provides a more balanced evaluation by penalizing both small and large errors equally making it useful for assessing the accuracy and performance consistency of the system. ITSE involves the integration of the product of the square of the error signal and time. This criterion places greater emphasis on errors that occur during the transient phase. Since time is included as a weighting factor, ITSE penalizes later errors more heavily, meaning the system is encouraged to correct errors quickly to minimize the overall performance value. It is particularly useful for evaluating systems where long-lasting errors are undesirable and faster stabilization is required. ITAE involves the integration of the product of the absolute value of the error signal and time. This criterion penalizes errors during the transient phase more heavily and evaluates the system in terms of stability and long-term performance. Since time is a weighting factor, ITAE encourages the system to minimize prolonged errors and achieve fast stabilization. It is particularly useful for assessing systems where overshoot and settling time need to be minimized for better overall performance.

Curve fitting method

Curve fitting is the process of expressing a continuous function that represents a set of discrete data points. Depending on the data errors, there are two main approaches to curve fitting: regression and interpolation.

Regression: Regression is used when the error rate is high, and the curve does not pass through each data point but rather represents the overall trend of the data. This method is often used for experimental data. One of the most widely used techniques in regression analysis is the least squares method. In this method, data collected from various applications in real-life scenarios are tabulated and analyzed to find a function that models the data. The goal is to determine the function that best fits the data table. The process of finding the function that best fits the data is called regression analysis. When the data contains significant errors, interpolation is not suitable as it may not provide satisfactory results for estimating intermediate values. In such cases, particularly for experimental data, the least squares method

provides better results. If the data can be represented by a straight line, the line fitting method is applied.

Interpolation: Interpolation is used when the error rate is low, and the curve is fitted to pass through each known discrete point. In this approach, the curve passes exactly through all given data points, ensuring a precise representation of the known values. Interpolation is generally used when data points are highly accurate and when estimating values between these points with minimal error is required. It is used to find the corresponding value of an intermediate point between known discrete points. The goal of interpolation is to determine an intermediate value $F(x)$ for any x between x_i and x_{i+1} , using the given measurements $f_0, f_1, f_2, \dots, f_n$ at the known points $x_0, x_1, x_2, \dots, x_n$. Linear interpolation is a useful method for curve fitting by using linear polynomials. It helps to generate new data points within the range of an already known set of discrete data points. Therefore, linear interpolation is the simplest method to estimate a value for a channel based on the given vector of predictions for that channel. In this study, the regression method was used for curve fitting.

Derivation of the sliding surface and control rule with relevant equations

In this part of the study, mathematical equations are derived to obtain the SMC parameters such as sliding surface, sliding surface slope, control rule for a second-order system controlled by the sliding mode control method. Thus, the relationship between the transfer function and the coefficients of the SMC controller to be designed is obtained in the case of controlling a second-order system with a known transfer function by SMC. In addition, the necessary design to obtain the optimal SMC parameters using optimization methods has been realized in this way.

The first step in SMC design involves determining the sliding surface. For this purpose, the commonly used equation is given in equation (5). Let $r(t)$ represent the reference input and $\omega(t)$ represent the desired output in the system, and let the tracking error be defined as $e(t) = r(t) - \omega(t)$. For an n th-order dynamic system, the linear sliding surface represents a stable dynamic system of order $(n - 1)$ and can generally be expressed as follows²:

$$S(x, t) = \left(\frac{d}{dt} + \lambda \right)^{n-1} e(t) = \left(\frac{d}{dt} + \lambda_{n-1} \right) \left(\frac{d}{dt} + \lambda_{n-2} \right) \dots \left(\frac{d}{dt} + \lambda_1 \right) e(t) \quad (5)$$

where λ represents the slope of the sliding surface, which determines the convergence rate of the system to the sliding mode. This equation forms the basis for defining the control rule in the SMC approach. The sliding surface is a function related to the tracking

error, which is the difference between the reference input and the measured output. Here, n represents the system's order, and λ is a positive number representing the slope of the sliding surface.

The control rule $u(t)$, which is the main objective of SMC, consists of the sum of two components:

$$u(t) = u_{eq}(t) + u_{sw}(t) \quad (6)$$

$u_{eq}(t)$ is the equivalent control ensuring motion along the sliding surface, and $u_{sw}(t)$ is the switching control ensuring state convergence.²

$$u_{sw}(t) = K \operatorname{sgn}(S) \quad (7)$$

Here, the $\operatorname{sgn}(S)$ function represents the signum function, and K is a positive constant that can be chosen large to suppress system uncertainties and unexpected dynamics. The switching control signal $u_{sw}(t)$ pulls the system states toward the sliding surface when $S(t) \neq 0$.

Once the system reaches the sliding surface, the switching control signal stops the switching process.

Due to this behavior, the equivalent control signal is continuous, whereas the switching control signal is discontinuous. The Lyapunov function is expressed by the equation as shown below.²

$$V = \frac{1}{2} S^2 \quad (8)$$

The commonly used system stability law for the sliding mode is given by equation (9),

$$S\dot{S} < 0 \quad \forall S \neq 0, V(0) = 0, V(S) > 0 \quad (9)$$

and is applied to derive equation (10) as follows:

$$\dot{V} = \frac{1}{2} \frac{d}{dt} S^2 \leq -\eta |S| \quad (10)$$

Here, S , $\dot{S}$, and η represent the sliding surface, the derivative of the sliding surface, and a constant value, respectively. The convergence condition is defined as;

$$S\dot{S} \leq -\eta |S|, S\dot{S} \leq -\eta \operatorname{sgn}(S)S \quad (11)$$

and the sliding mode condition is defined in equation (12) as follows²:

$$\dot{S} \operatorname{sgn}(S) \leq -\eta \quad (12)$$

If the condition $\eta > 0$ is satisfied, the system enters the sliding mode. Although traditional SMC offers advantages such as robustness, it also has a notable drawback known as chattering. The chattering phenomenon arises from high-frequency switching and the sharpness of the switching function. While its effects are not highly noticeable in simulation studies, it becomes a

significant issue in real-time applications. In real-time implementations, the infinite fast switching assumption, which is the foundation of general SMC theory, cannot be achieved due to time delays and physical limitations. Therefore, minimizing chattering becomes crucial for practical applications.

For a second-order system, the sliding surface can be expressed as shown below:

$$S(x, t) = \left(\frac{d}{dt} + \lambda \right)^{2-1} e(t) = \left(\frac{d}{dt} + \lambda \right) e(t) \quad (13)$$

$$= \dot{e}(t) + \lambda e(t)$$

$S(t)$ is the sliding surface, $e(t)$ is the tracking error, defined as the difference between the reference input and the system output, λ is a positive constant representing the slope of the sliding surface, which determines the convergence rate of the system toward the desired output.

If the transfer function of a second-order system to be controlled using the SMC method is assumed in the frequency domain as follows, where $U(s)$ represents the control rule of SMC and serves as the input to the transfer function, and $W(s)$ represents the output of the transfer function, the transfer function, and control rule can be expressed in the frequency domain as shown in equations (14) and (15).³⁶

$$\frac{W(s)}{U(s)} = \frac{1}{(As^2 + Bs + C)} \quad (14)$$

One can write equation (14) like as equation (15)

$$U(s) = AW(s)s^2 + BW(s)s + CW(s) \quad (15)$$

Thus, the control rule in the time domain can be expressed as shown below;

$$u(t) = A\ddot{w}(t) + B\dot{w}(t) + Cw(t) \quad (16)$$

For a second-order system, the derivative of the sliding surface in the time domain is expressed as;

$$\dot{S} = (\dot{e}(t) + \lambda e(t))' = \ddot{e}(t) + \lambda \dot{e}(t) = -K.Sig(n(S)) \quad (17)$$

If the difference between the reference input $r(t)$ and the SMC system output $\omega(t)$ is considered as the error $e(t)$, the first and second derivatives of the error can be expressed as shown in equations (18)–(20).³⁶

$$e(t) = r(t) - \omega(t) \quad (18)$$

$$\dot{e}(t) = \dot{r}(t) - \dot{\omega}(t) \quad (19)$$

$$\ddot{e}(t) = \ddot{r}(t) - \ddot{\omega}(t) \quad (20)$$

If equations (18)–(20) are substituted into equation (17), equation (21) is obtained as follows:

$$\dot{S} = \ddot{r}(t) - \ddot{\omega}(t) + \lambda \dot{r}(t) - \lambda \dot{\omega}(t) = -K.Sig(n(S)) \quad (21)$$

Since the reference signal is a constant value, its first and second derivatives will be zero. Therefore, by simplifying equation (21), equation (22) is obtained. The second derivative of the SMC system output is given in equation (22) as follows:

$$\ddot{\omega}(t) = K.Sig(n(S)) - \lambda \dot{\omega}(t) \quad (22)$$

When equation (22) is substituted into equation (16), the control rule is obtained as equation (23). With the mathematical equation derived, the design values for a system to be controlled using the SMC method can be implemented through its transfer function using the MATLAB Simulink platform, as shown in Figure 2. In the SMC method, the slope of the sliding surface is expressed as λ , similarly the signum function coefficient is denoted as K in both the equation and the Simulink control diagram. The primary goal in the SMC method is to determine the optimum values of K and λ to achieve robust and effective control.

$$u(t) = A.K.Sig(n(S)) - A.\lambda.\dot{\omega}(t) + B\dot{\omega}(t) + C\omega(t) \quad (23)$$

In this paper, a method is presented for design of SMC by using a reference model approach. For this purpose, the parameters of the SMC controller are found using the Fmincon optimization algorithm method and Integral Performance criteria. The control diagram of the system controlled by the SMC method based on the reference model is as shown in Figure 3.

By applying a step function simultaneously from the input, the reference system with a second-order transfer function (given in equation (24)) and the second-order system controlled by the SMC method (given in equation (25)) were compared by calculating the difference between their outputs. As shown in Figure 4, the design for calculating the error was implemented in the MATLAB m-file and Simulink platform to obtain the ISE, IAE, ITSE, and ITAE values. In this study, the selected transfer functions, the corresponding SMC parameter values, and the calculated performance criteria are presented in tables.

$$H(s) = \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)} \quad (24)$$

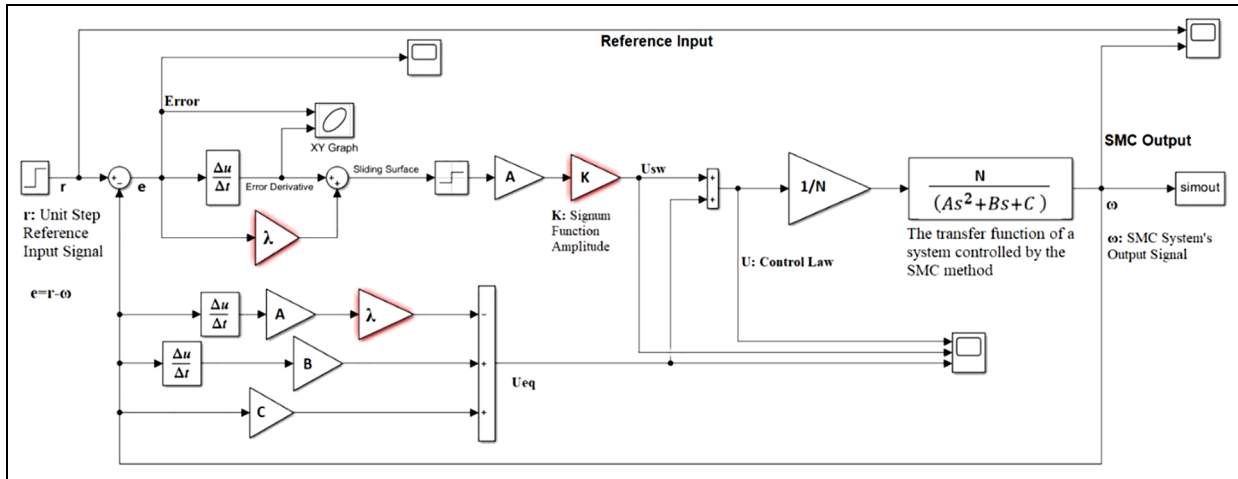


Figure 2. Design parameters of a system controlled by the SMC method.

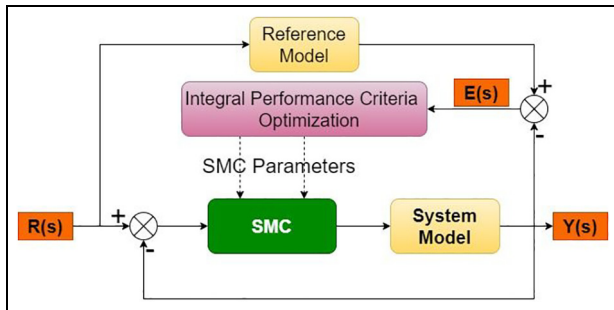


Figure 3. Block diagram of the presented method.

$$G(s) = \frac{1}{(As^2 + Bs + C)} \quad (25)$$

The reference system, denoted by equation (24), has a second-order transfer function. If we express the parameters of this transfer function as follows: ζ represents the damping ratio, ω_n represents the natural frequency of the system in rad/s, in other words, ω_n indicates the frequency at which the system will oscillate in the absence of an external force. The poles of the reference system can be calculated using equation (26) as follows:

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} \quad (26)$$

The damping ratio ζ determines the stability and response characteristics of a second-order system. The system's behavior varies based on the value of ζ as follows³⁷:

- $\zeta < 0$: Unstable System
- $\zeta = 0$: Pure Oscillation system
- $0 < \zeta < 1$: Underdamped System (Oscillatory Response)
- The system exhibits oscillations before reaching equilibrium.

- $\zeta = 1$: Critically Damped System (Non-Oscillatory Response)
- The system returns to equilibrium without oscillating and in the shortest possible time.
- $\zeta > 1$: Overdamped System (Slow Non-Oscillatory Response)
- The system does not oscillate but takes a longer time to settle to equilibrium.

The ζ significantly affects the rise time, settling time, and oscillation behavior of the system.

In the reference system described by equation (24), the ω_n is kept constant at 1 rad/s, while the ζ varies between 0.5 and 1, as shown in Tables 1 to 4. Using equation (26), the roots of the reference system for selected damping ratio values ζ have been calculated and presented in Tables 1 to 4. In the analysis conducted for all roots, it has been observed that the real parts of all roots are negative. Therefore, it can be concluded that the system will reach equilibrium over time. The system is stable and will gradually settle to a steady-state. In this study, while the natural frequency of the reference system was kept constant at 1 rad/s, the damping ratio of the reference system was varied between 0.5 and 1. During this process, the transfer function of the system controlled by SMC was kept constant, and the differences between the reference system output and the SMC system output were calculated to obtain the ISE, IAE, ITSE, and ITAE error performance values.

Research findings

In this study, a second-order reference system with a transfer function and a second-order system controlled by the SMC method were simultaneously subjected to a unit step function in the Simulink platform. The aim was to equalize the difference between the output of the SMC-controlled system and the reference system to zero. To achieve this, the initial values of λ and K were

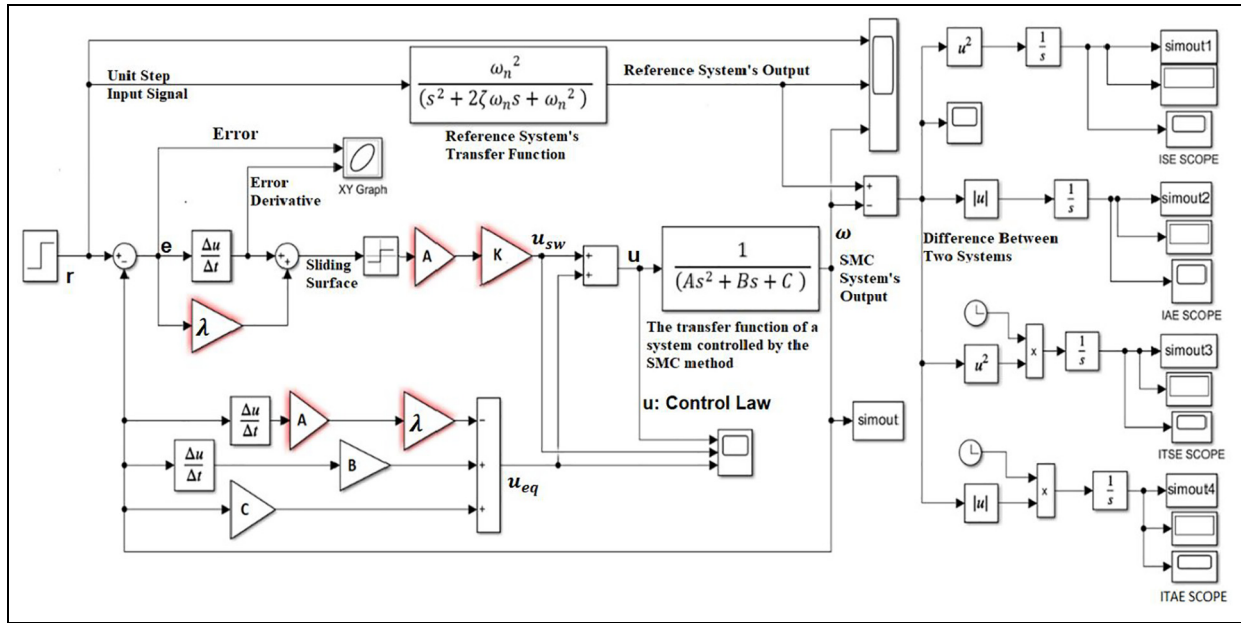


Figure 4. Design parameters for calculating the error difference between outputs of the system controlled by the SMC method and the reference system.

Table 1. The optimum SMC parameters calculated using Fmincon for the reference system with a given transfer function and the system controlled with SMC, the ISE values, and the SMC parameters obtained using curve fitting.

Reference Model Damping Ratio (ζ)	A = 1, B = 4, C = 5, λ , K			Values obtained by using the Fmincon optimization algorithm		Values obtained by using the curve fitting method	
	λ	K	ISE	Curve fitting λ	Curve fitting K		
0.5	0.9999	1.0000	0.0943	0.8710	1.9721		
0.55	0.9363	3.4458	0.1004	0.9363	1.8191		
0.56	0.9990	10.0000	0.0569	0.9493	1.7885		
0.57	1.8236	1.2549	0.0326	0.9624	1.7580		
0.58	0.7331	47.9847	0.1019	0.9755	1.7274		
0.59	1.1032	1.3622	0.0481	0.9885	1.6968		
0.6	0.7525	41.9266	0.0948	1.0016	1.6662		
0.61	1.1836	1.1923	0.0348	1.0146	1.6356		
0.62	6.9731	3.2508	0.0272	1.0277	1.6050		
0.63	1.0776	1.3086	0.0383	1.0408	1.5744		
0.64	0.8048	2.6336	0.0628	1.0538	1.5439		
0.65	0.7954	2.6983	0.0611	1.0669	1.5133		
0.7	0.9549	1.1189	0.0227	1.1322	1.3604		
0.71	0.9958	1.4819	0.0395	1.1452	1.3298		
0.72	0.8126	1.0061	0.0217	1.1583	1.2992		
0.73	2.9473	1.2686	0.0047	1.1713	1.2686		
0.74	0.8002	1.3711	0.0274	1.1844	1.2380		
0.75	0.5491	49.1689	0.0594	1.1975	1.2074		
0.76	0.9089	0.9149	0.0112	1.2105	1.1768		
0.8	1.4977	0.7824	0.0044	1.2627	1.0545		
0.81	0.9999	0.6971	0.0033	1.2758	1.0239		
0.82	1.0512	0.6842	0.0029	1.2889	0.9933		
0.83	1.0980	0.6772	0.0030	1.3019	0.9627		
0.84	1.0371	0.6755	0.0028	1.3150	0.9322		
0.85	34.2219	10.5225	0.0072	1.3280	0.9016		
0.9	54.2369	15.3419	0.0106	1.3933	0.7487		
0.93	1.4325	0.6555	0.0111	1.4325	0.6569		
0.95	54.3751	15.2102	0.0122	1.4586	0.5957		
1	8.2378	2.0847	0.0266	1.5239	0.4428		

Table 2. The optimum SMC parameters calculated using Fmincon for the reference system with a given transfer function and the system controlled with SMC, the IAE values, and the SMC parameters obtained using curve fitting.

Reference Model Damping Ratio (ζ)	$A = 1, B = 4, C = 5, \lambda, K$				
	Values obtained using the Fmincon optimization algorithm			Values obtained using the curve fitting method	
	λ	K	IAE	Curve fitting λ	Curve fitting K
0.5	1.2269	1.2161	0.6109	1.1723	0.8779
0.55	1.3063	0.8755	0.4809	1.3063	0.8755
0.6	1.4059	0.8722	0.3569	1.4403	0.8731
0.65	1.5034	0.7630	0.3714	1.5743	0.8707
0.7	2.1448	1.0951	0.1857	1.7083	0.8683
0.72	1.5411	0.8598	0.1770	1.7619	0.8673
0.75	2.7928	1.2590	0.1679	1.8424	0.8659
0.8	1.5549	0.7705	0.1507	1.9764	0.8635
0.81	2.0104	0.9929	0.1872	2.0032	0.8630
0.82	1.3472	0.6764	0.1495	2.0300	0.8625
0.83	2.7477	1.1481	0.1668	2.0568	0.8620
0.84	1.8351	0.7870	0.1700	2.0836	0.8616
0.85	1.7177	0.7512	0.1717	2.1104	0.8611
0.87	1.6897	0.7511	0.1811	2.1640	0.8601
0.88	6.6660	2.2709	0.1799	2.1908	0.8596
0.89	59.5393	25.5873	0.6111	2.2176	0.8592
0.9	62.0051	27.5805	0.6679	2.2444	0.8587
0.92	2.0491	0.8578	0.2387	2.2980	0.8577
0.94	2.1139	0.8271	0.2529	2.3516	0.8568
0.95	2.3784	0.9091	0.2643	2.3784	0.8563
1	3.3116	0.7709	0.5788	2.5124	0.8539

determined, and through iteration steps, the λ and K values that minimized the error between the outputs of the two systems were obtained. Subsequently, optimization methods were applied to minimize the error values. To achieve the minimum error, different error calculation methods such as ISE, IAE, ITSE, and ITAE were used to find the respective error values.

Optimization was carried out using the Fmincon tool in the Matlab platform. The optimization was performed using fmincon in MATLAB with the aim of minimizing the output error between the reference and SMC-controlled systems. The optimization variables were λ (sliding surface slope) and K (switching gain), which were bounded within the range [0, 100]. Initial values for these parameters were selected based on prior simulations to ensure stable operation and convergence. Typically, the initial values for the optimization variables were set as $\lambda = 1$ and $K = 1$ based on prior simulation results, which ensured stable initial performance and numerical convergence.

$A = 1, B = 4, C = 5$ for the SMC controlled system in (25) and $\omega_n = 1$ is fixed for the reference system in (24) and ζ is varied between 0.5 and 1. The optimized SMC parameters (λ and K) for different ζ values are obtained and presented in Tables 1 to 4. The relationship between and the optimized parameters is modeled using linear regression techniques. The obtained results were presented in this paper.

The SMC parameters obtained by optimizing the error between the reference system and the output of the system controlled by SMC and integral

performance criteria (ISE, IAE, ITSE and ITAE error performance) values are also given in Table 1 to Table 4 respectively. The plot of λ and K values against ζ obtained by using the Fmincon optimization algorithm and the linear polynomial of Curve Fitter Toolbox while varying ζ between 0.5 and 1 is shown in Figures 5 to 12. Additionally, Table 1 displays the SMC parameter values determined using the linear polynomial of the curve fitting approach as well as the SMC parameters obtained during the ISE calculation utilizing the optimization method.

As seen in Table 1, the lowest ISE error value for the output differences of the specified systems has been obtained when $\zeta = 0.84$.

$$\lambda = 1,3058\zeta + 0,2181 \quad (27)$$

When Figures 5 and 6 are examined, it is observed that the λ , SMC design parameter obtained using Fmincon does not exhibit significant fluctuations in the range of $\zeta = 0.5$ to $\zeta = 0.84$. The data obtained through the Fmincon optimization method and the relationship between λ and K , derived using the linear polynomial of the Curve Fitter Toolbox, with ζ , are presented in equations (27) and (28).

$$K = -3,0585\zeta + 3,5013 \quad (28)$$

Table 2 displays the SMC parameter values determined using the linear polynomial of the curve fitting approach as well as the SMC parameters obtained

Table 3. The optimum SMC parameters calculated using Fmincon for the reference system with a given transfer function and the system controlled with SMC, the ITSE values, and the SMC parameters obtained using curve fitting.

Reference Model Damping Ratio (ζ)	$A = 1, B = 4, C = 5, \lambda, K$				
	Values obtained using the Fmincon optimization algorithm			Values obtained using the curve fitting method	
	λ	K	ITSE	Curve fitting λ	Curve fitting K
0.50	1.8695	2.7283	0.3106	0.8712	1.7016
0.55	1.9941	1.9920	0.2047	0.9598	1.5951
0.60	1.1268	1.4886	0.1311	1.0485	1.4886
0.65	1.3709	1.1943	0.0733	1.1372	1.3822
0.70	1.2259	1.4197	0.0832	1.2259	1.2757
0.75	1.1692	1.1692	0.0555	1.3146	1.1693
0.76	1.2692	1.2664	0.0753	1.3324	1.1480
0.77	1.0641	1.1749	0.0515	1.3501	1.1267
0.78	1.0032	1.0039	0.0333	1.3678	1.1054
0.79	0.8675	0.9023	0.0286	1.3856	1.0841
0.80	0.6733	31.7691	0.0808	1.4033	1.0628
0.81	0.9999	1.0001	0.0337	1.4211	1.0415
0.82	0.8352	1.0106	0.0285	1.4388	1.0202
0.83	0.8012	0.9799	0.0265	1.4565	0.9989
0.84	0.8127	0.7842	0.0174	1.4743	0.9777
0.85	0.8613	0.7791	0.0129	1.4920	0.9564
0.88	2.6848	0.8612	0.0406	1.5452	0.8925
0.90	1.5824	0.7430	0.0299	1.5807	0.8499
0.91	2.6874	0.8819	0.0346	1.5984	0.8286
0.93	60.8047	16.8948	0.0329	1.6339	0.7860
0.94	10.2060	2.6848	0.0547	1.6517	0.7647
0.95	66.3212	18.0487	0.0379	1.6694	0.7435
1.00	11.4140	3.0090	0.0554	1.7581	0.6370

Table 4. The optimum SMC parameters calculated using Fmincon for the reference system with a given transfer function and the system controlled with SMC, the ITAE values, and the SMC parameters obtained using curve fitting.

Reference Model Damping Ratio (ζ)	$A = 1, B = 4, C = 5, \lambda, K$				
	Values obtained using the Fmincon optimization algorithm			Values obtained using the curve fitting method	
	λ	K	ITAE	Curve fitting λ	Curve fitting K
0.50	1.8825	1.4109	2.1791	1.7434	0.9784
0.51	56.7315	31.7641	2.3038	1.7312	0.9680
0.52	22.6529	18.4581	2.4880	1.7190	0.9576
0.53	1.1794	0.8249	2.2029	1.7068	0.9472
0.54	1.1789	0.8249	2.0956	1.6946	0.9368
0.55	2.0535	1.3087	1.6530	1.6824	0.9264
0.56	3.4309	1.8635	1.5900	1.6702	0.9160
0.57	1.1855	0.8215	1.8097	1.6580	0.9057
0.58	1.6638	1.0744	1.4167	1.6458	0.8953
0.59	1.4482	0.8209	1.5783	1.6336	0.8849
0.60	1.1789	0.8249	1.5450	1.6214	0.8745
0.65	1.2422	0.8026	1.1572	1.5604	0.8225
0.70	1.2381	0.7706	0.8689	1.4994	0.7706
0.75	1.3545	0.6468	0.8363	1.4384	0.7186
0.80	1.3551	0.6454	0.5074	1.3774	0.6666
0.81	1.3545	0.6449	0.4604	1.3652	0.6562
0.82	1.3530	0.6459	0.4236	1.3530	0.6459
0.83	1.3531	0.6454	0.3986	1.3408	0.6355
0.84	1.3525	0.6454	0.3816	1.3286	0.6251
0.85	1.3522	0.6451	0.3729	1.3164	0.6147
0.90	1.4053	0.6034	0.4362	1.2554	0.5627
0.92	1.2310	0.4900	0.4678	1.2310	0.5419
0.95	1.0567	0.5476	0.4067	1.1944	0.5108
0.96	1.1043	0.4169	0.6165	1.1822	0.5004
1.00	1.5992	0.5520	0.9709	1.1334	0.4588

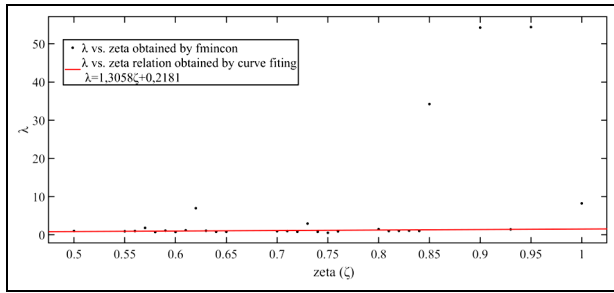


Figure 5. The slope values of the sliding surface obtained through optimization for the SMC system and the slope values obtained using curve fitting, based on the damping ratio of the reference system using ISE.

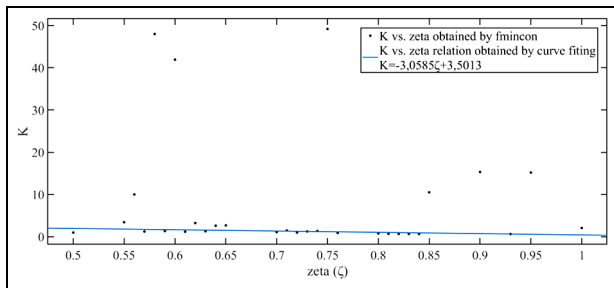


Figure 6. The signum function coefficient values obtained through optimization for the SMC system and the signum function coefficient values obtained using curve fitting, based on the damping ratio of the reference system using ISE.

during the IAE calculation utilizing the optimization method.

When Figures 7 and 8 are examined, it is observed that the λ and K SMC design parameters obtained using Fmincon do not exhibit significant fluctuations in the range of $\zeta = 0.5$ to $\zeta = 0.87$. The data obtained through the Fmincon optimization method and the relationship between λ and K derived using the linear polynomial of the Curve Fitter Toolbox, with ζ , are presented in equations (29) and (30) respectively.

$$\lambda = 2,6802\zeta - 0,1678 \quad (29)$$

$$K = -0,0479\zeta + 0,9018 \quad (30)$$

Table 3 displays the SMC parameter values determined using the linear polynomial of the curve fitting approach as well as the SMC parameters obtained during the ITSE calculation utilizing the optimization method. As seen in Table 3, the lowest ITSE error performance value for the output differences of the specified systems was obtained when $\zeta = 0.85$.

When Table 3, Figures 9 and 10 are examined, it is observed that among the SMC design parameters obtained using Fmincon, the λ value does not exhibit significant fluctuations in the range of $\zeta = 0.5$ to

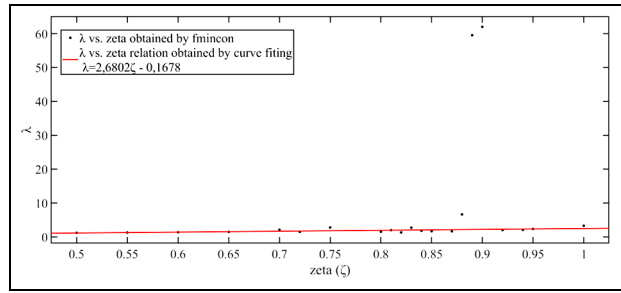


Figure 7. The slope values of the sliding surface obtained through optimization for the SMC system and the slope values obtained using curve fitting, based on the damping ratio of the reference system using IAE.

$\zeta = 0.9$, while the K value remains relatively stable in the range of $\zeta = 0.5$ to $\zeta = 0.79$. The data obtained through the Fmincon optimization method and the relationship between λ and K , derived using the linear polynomial of the Curve Fitter Toolbox, with ζ , are presented in equations (31) and (32).

$$\lambda = 1,7739\zeta - 0,0158 \quad (31)$$

$$K = -2,1291\zeta + 2,7661 \quad (32)$$

Table 4 displays the SMC parameter values determined using the linear polynomial of the curve fitting approach as well as the SMC parameters obtained during the ITAE calculation utilizing the optimization method. As seen in Table 4, the lowest ITAE error performance value for the output differences of the specified systems was obtained when $\zeta = 0.85$.

The transfer function values specified in Table 4 and the corresponding ζ values were analyzed along with the optimized λ and K SMC parameters and the resulting error calculations. Based on the variation of ζ , the graphs of λ and K SMC parameters were obtained separately, as shown in Figures 11 and 12. When Table 4 is examined, it is observed that the λ and K SMC design parameters obtained through the Fmincon optimization algorithm do not exhibit significant fluctuations in the range of

$\zeta = 0.53$ to $\zeta = 1$. Using the curve fitting method, the relationship between the SMC design parameters λ and K and the ζ value is expressed by equations (33) and (34) respectively.

$$\lambda = -1,2200\zeta + 2,3534 \quad (33)$$

$$K = -1,0392\zeta + 1,4980 \quad (34)$$

The values from Tables 1 to 4 have been analyzed. When examining the error performance values from the tables, it was observed that the lowest ISE value was obtained at $\zeta = 0.84$, the lowest IAE value at $\zeta = 0.82$, the lowest ITSE value at $\zeta = 0.85$, and the

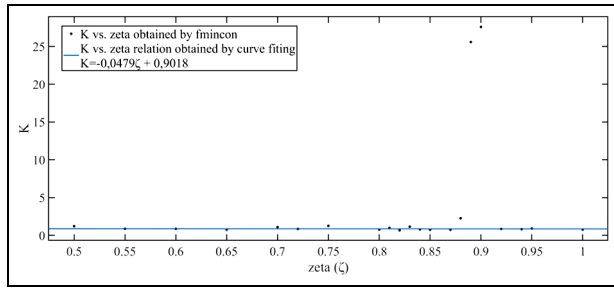


Figure 8. The signum function coefficient values obtained through optimization for the SMC system and the signum function coefficient values obtained using curve fitting, based on the damping ratio of the reference system using IAE.

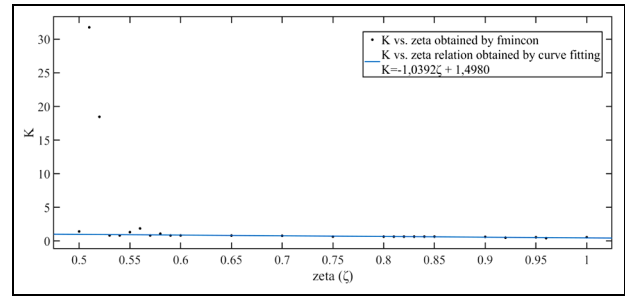


Figure 12. The signum function coefficient values obtained through optimization for the SMC system and the signum function coefficient values obtained using curve fitting, based on the damping ratio of the reference system using ITAE.

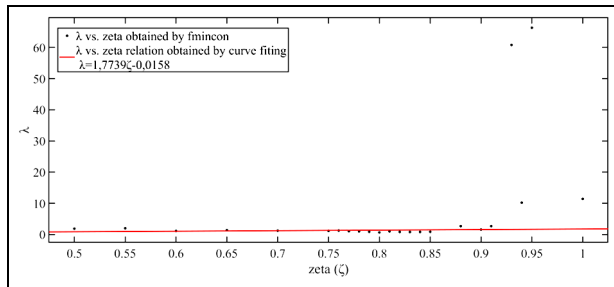


Figure 9. The slope values of the sliding surface obtained through optimization for the SMC system and the slope values obtained using curve fitting, based on the damping ratio of the reference system using ITSE.

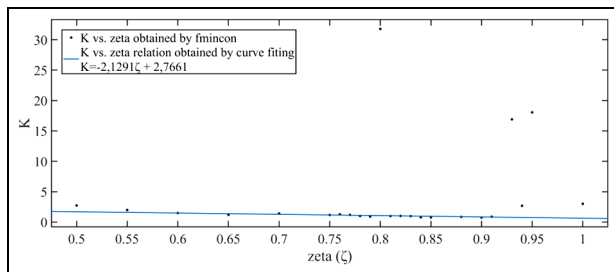


Figure 10. The signum function coefficient values obtained through optimization for the SMC system and the signum function coefficient values obtained using curve fitting, based on the damping ratio of the reference system using ITSE.

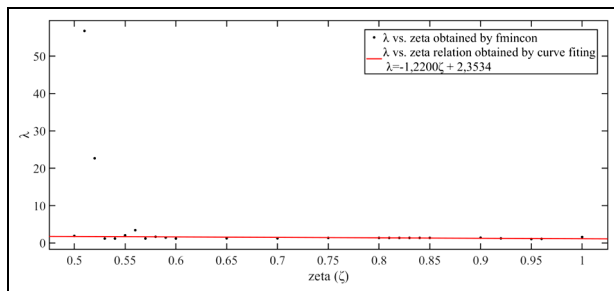


Figure 11. The slope values of the sliding surface obtained through optimization for the SMC system and the slope values obtained using curve fitting, based on the damping ratio of the reference system using ITAE.

lowest ITAE value at $\zeta = 0.85$. These values are presented in Table 5. Additionally, the sliding surface graphs and the output graphs of the system controlled with SMC, corresponding to $\zeta = 0.82$, $\zeta = 0.84$, and $\zeta = 0.85$, in relation to the ISE, IAE, ITSE, and ITAE error performance values, are shown in Figures 13 to 18. Figures 13 to 18 compare the step responses of the reference system and the optimized SMC controlled system, confirming improved response times and reduced error. When all the tables are examined, it is observed that the lowest error was obtained with ISE at $\zeta = 0.84$.

To demonstrate the impact of optimization on chattering suppression, the switching control signal was analyzed before and after optimizing λ and K were optimized. Figures 19 and 20 present the control signals under non-optimized and optimized conditions, respectively. The optimized case is seen to exhibit significantly reduced oscillation amplitude and frequency in the switching control, confirming the effectiveness of the proposed optimization in mitigating the chattering phenomenon.

While the proposed optimization framework yields promising results in simulation, several limitations should be noted. Firstly, the approach assumes an accurately modeled second-order linear system, so extending it to nonlinear or time-varying systems would require further reformulation. Secondly, the results are based entirely on MATLAB/Simulink simulations, and no real-time or hardware-based validation has been conducted to assess the feasibility of implementation. Thirdly, the performance of the fmincon algorithm may depend on the initial guesses for the parameters and the solver settings, which can affect convergence in more complex systems. Future studies should address these factors to enhance robustness and practical applicability.

Discussions

i. Main findings:

- The proposed reference model-based optimization framework effectively determined the

Table 5. The optimum ISE, IAE, ITSE, and ITAE values calculated using Fmincon for the error between the Reference System and the SMC-controlled system at $\zeta = 0.82$, $\zeta = 0.84$, and $\zeta = 0.85$.

$\zeta = 0.82$	$\lambda = 1.0512$ $K = 0.6842$ ISE = 0.0029	$\lambda = 1.3472$ $K = 0.6764$ IAE = 0.1495	$\lambda = 0.8352$ $K = 1.0106$ ITSE = 0.0285	$\lambda = 1.3530$ $K = 0.6459$ ITAE = 0.4236
$\zeta = 0.84$	$\lambda = 1.0371$ $K = 0.6755$ ISE = 0.0028	$\lambda = 1.78351$ $K = 0.7870$ IAE = 0.1700	$\lambda = 0.8127$ $K = 0.7842$ ITSE = 0.0174	$\lambda = 1.3525$ $K = 0.6454$ ITAE = 0.3816
$\zeta = 0.85$	$\lambda = 34.2219$ $K = 10.5225$ ISE = 0.0072	$\lambda = 1.7177$ $K = 0.7512$ IAE = 0.1717	$\lambda = 0.8613$ $K = 0.7791$ ITSE = 0.0129	$\lambda = 1.3522$ $K = 0.6451$ ITAE = 0.3729

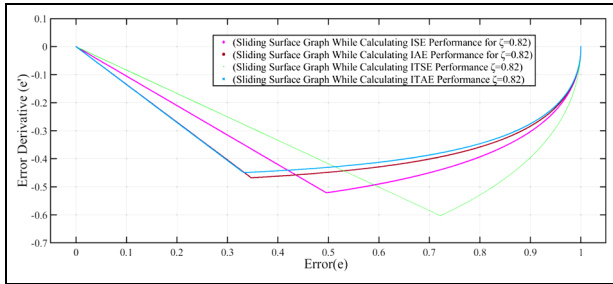


Figure 13. Sliding surfaces of the SMC system for $\zeta = 0.82$.

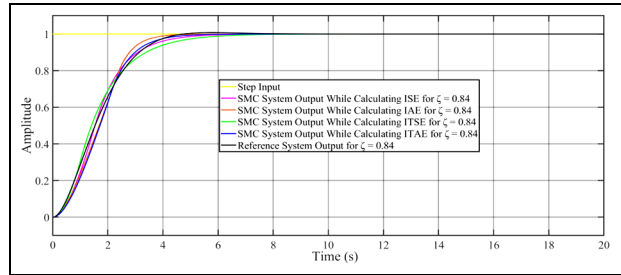


Figure 16. Output signals of the SMC system for $\zeta = 0.84$.

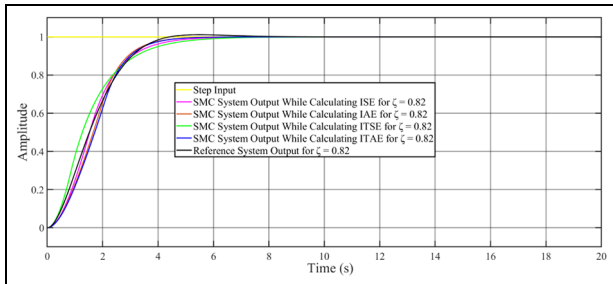


Figure 14. Output signals of the SMC system for $\zeta = 0.82$.

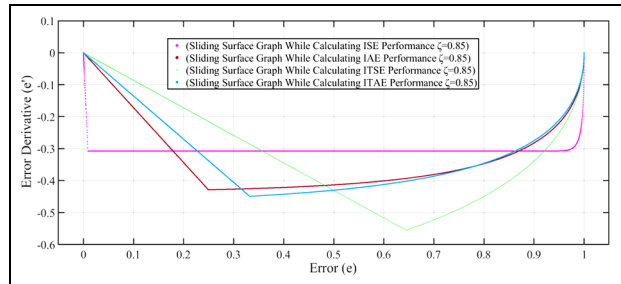


Figure 17. Sliding surfaces of the SMC system for $\zeta = 0.85$.

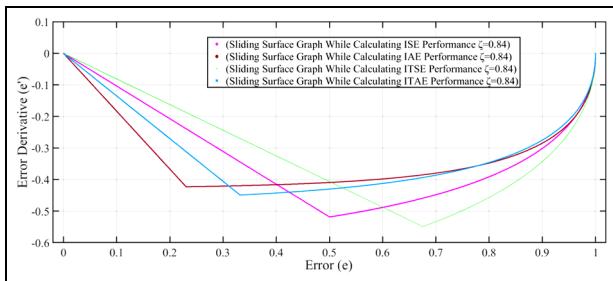


Figure 15. Sliding surfaces of the SMC system for $\zeta = 0.84$.

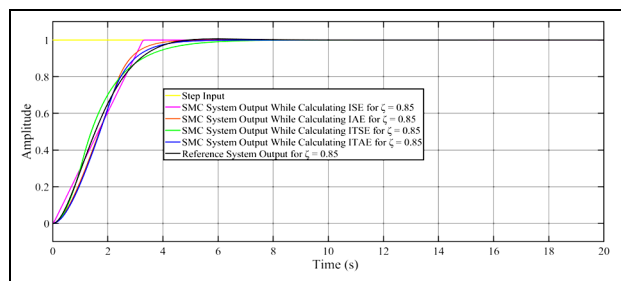


Figure 18. Output signals of the SMC System for $\zeta = 0.85$.

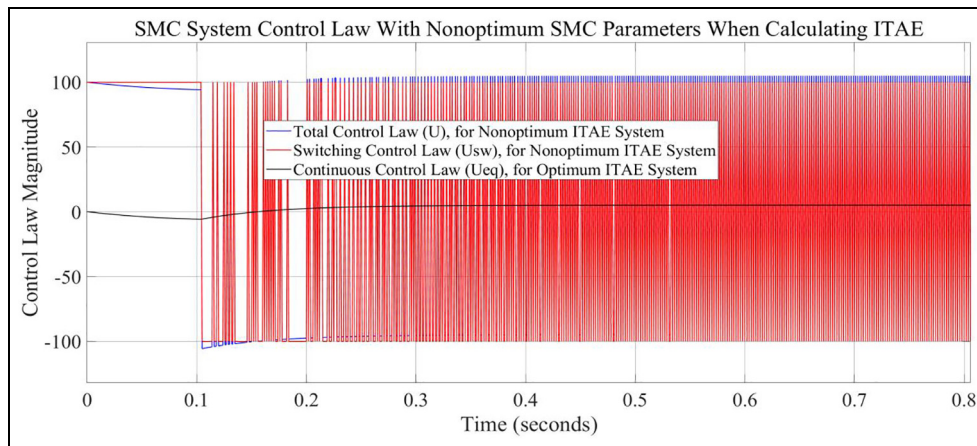


Figure 19. Switching control signal before optimization of λ and K (non-optimized case).

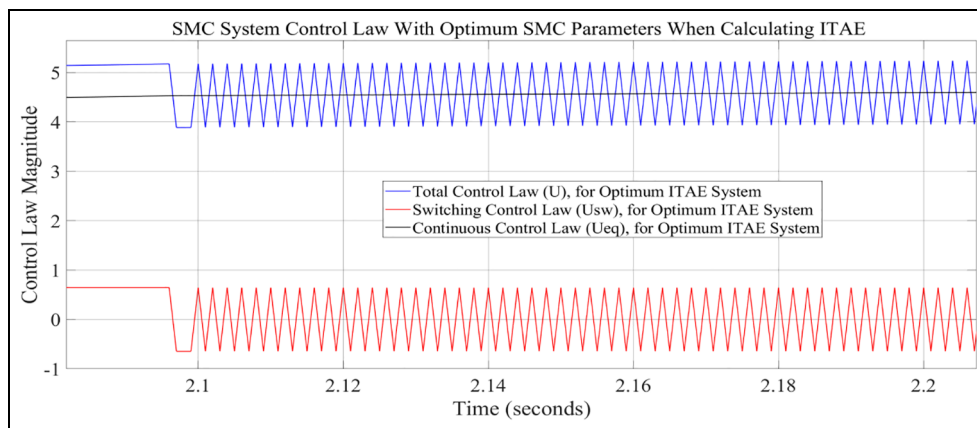


Figure 20. Switching control signal after optimization of λ and K (optimized case).

optimal SMC parameters (λ and K) that minimized error performance criteria.

- The curve-fitting model accurately generalized the relationship between damping ratio and SMC parameters, reducing the need for repeated simulations.

ii. Comparison with other studies:

- Compared to traditional trial-and-error methods or fixed-rule SMC tuning approaches, our method provides a systematic, repeatable, and data-driven solution.
- The use of integral criteria and regression modeling offers greater adaptability, similar to recent intelligent control frameworks cited in the recent literature (e.g. neural networks and fuzzy logic).

iii. Implication and explanation:

- The optimized parameters improved time-domain performance (settling time, overshoot and tracking accuracy) and significantly reduced chattering.
- Analytical models from curve fitting enabled controller tuning for unseen damping values, supporting efficient and scalable control design.

iv. Strengths and limitations:

- **Strengths:** Clear optimization strategy, validated performance across four error metrics, analytical parameter model, and visual confirmation of chattering reduction.
- **Limitations:** No experimental (hardware) implementation was included; the approach is currently limited to second-order linear systems.

v. Recommendation, and future direction:

- The method is an effective tool for systematically tuning SMC controllers systematically and may be extended to nonlinear or higher-order systems.
- Future studies will focus on extending the method to nonlinear systems and implementing it in real-time embedded control environments.

Conclusions

This study presents a systematic approach for optimizing SMC controller parameters based on a second-order reference model. The proposed method efficiently determines the optimal sliding surface slope λ and

control gain K by leveraging the Fmincon optimization algorithm, integral performance criteria, and curve fitting techniques. By establishing a direct relationship between the damping ratio ζ and the optimized parameters, the need for extensive simulations is minimized. Key findings indicate that the optimization method effectively reduces error values across all considered criteria while enhancing system performance. The study also highlights that selecting appropriate λ and K values significantly improves system response and mitigates the chattering effect. Additionally, the Curve Fitter Toolbox was used to further refine the relationship between the optimized parameters and the damping ratio, enabling precise interpolation of parameter values for untested scenarios. This approach provides a more flexible and computationally efficient method for parameter tuning in various control applications.

Compared to conventional SMC tuning methods, which often rely on manual adjustments and extensive trial-and-error, the proposed optimization-based approach offers a more structured, and efficient alternative. Although the initial optimization process involves a moderate computational cost due to simulation-based evaluations, this is significantly outweighed by the advantages of automation, consistency, and greater accuracy in parameter selection. Moreover, the use of curve fitting after optimization allows for rapid parameter estimation for untested damping ratios, further reducing the need for repeated computation in future implementations. One of the current limitations of this study is the lack of experimental validation through hardware implementation. Although the simulation results demonstrate the effectiveness of the proposed optimization method, it is necessary to deploy it in real-time environments is necessary to evaluate its robustness under physical constraints and noise. Future work will therefore focus on implementing the optimized SMC controller on embedded platforms, such as dSPACE, Arduino, or STM32-based systems in order to assess its performance in real-world conditions.

Future research should explore adaptive SMC strategies and intelligent control techniques such as fuzzy logic or neural networks to further enhance performance. Additionally, experimental validation in real-world applications would provide valuable insights into practical implementation challenges. Additionally, future research could examine the application of this optimization framework to systems with time-varying parameters by integrating adaptive or gain-scheduled SMC structures to guarantee robustness and responsiveness under dynamic operating conditions. Although this study focuses on second-order systems, the proposed optimization framework can be adapted to higher-order and non-linear systems. For higher-order systems, the sliding surface and control law can be extended using the higher-order derivatives of the error signal. Similarly, in the case of nonlinear systems, as long as a differentiable system model is available, the

same optimization strategies can be employed. Future studies could extend this method to nonlinear dynamic systems or systems with uncertain parameters, thereby broadening the scope and applicability of the proposed approach. While the current study is limited to simulation-based evaluations, future work will focus on real-time implementation and hardware-in-the-loop testing. This will further validate the practical applicability and robustness of the optimized SMC parameters in real-world conditions.

Acknowledgement

We thank Turkish Academy of Sciences (TÜBA) for supporting this work.

Authors' Note

The authors contributed equally to this publication.

ORCID iD

İbrahim Halil Teke  <https://orcid.org/0000-0003-1007-3636>

Ethics approval

Ethical approval was not obtained for this study as it was not required.

Informed consent/Consent to participate

Informed consent was not required for this study.
Not applicable.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability statement

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

References

1. Kiss JM, Szemes PT and Aradi P. Sliding mode control of a servo system in LabVIEW: comparing different control methods. *Int Rev Appl Sci Eng* 2021; 12: 201–210.
2. Durdu A and Dursun EH. Sliding mode control for position tracking of servo system with a variable loaded DC motor. *Elektron Elektrotech* 2019; 25: 8–16.
3. Naik AK, Kar SK and Sahu BK. Speed control of DC motor using linear and non-linear controllers. In: *2021 1st Odisha international conference on electrical power engineering, communication and computing technology (ODICON)*. Bhubaneswar, India: IEEE, 2021, pp.1–5.

4. Howimanporn S, Chookaew S and Silawatchananai C. Comparison between PID and sliding mode controllers for rotary inverted pendulum using PLC. In: *2020 4th International conference on automation, control and robots (ICACR)*. Rome, Italy: IEEE, 2020, pp.122–126.
5. Jeong SK, Kim BR and Park KB. Terminal sliding mode control system with minimum convergence time for DC motor systems. In: *2012 12th international conference on control, automation and systems*. Jeju, Korea (South): IEEE, 2012, pp.55–58.
6. Santacruz D and Camacho O. PID based on sliding mode control design from a reduced order model: a practical evaluation. In: *2022 IEEE biennial congress of Argentina (ARGENCON)*. San Juan, Argentina: IEEE, 2022, pp.1–6.
7. Rakhonde S and Kulkarni V. Sliding mode controller (SMC) governed speed control of DC motor. In: *2018 3rd IEEE International conference on recent trends in electronics, information & communication technology (RTEICT)*. Bangalore, India: IEEE, 2018, pp.1657–1662.
8. Zarour L, Abed K, Hacil M, et al. Control and optimisation of photovoltaic water pumping system using sliding mode. *Bull Pol Acad Sci Tech Sci* 2019; 67: 605–611.
9. Suman K and Mathew AT. Speed control of permanent magnet synchronous motor drive system using PI, PID, SMC and SMC plus PID controller. In: *2018 international conference on advances in computing, communications and informatics (ICACCI)*. Bangalore, India: IEEE, 2018, pp.543–549.
10. Balaji M, Sarangi SK and Pattnaik M. Design of a DC motor based wind turbine emulator using sliding mode control approach. In: *2019 IEEE 1st international conference on energy, systems and information processing (ICE-SIP)*. Chennai, India: IEEE, 2019, pp. 1–5.
11. Amirkhani S, Mobayen S, Ilaee N, et al. Fast terminal sliding mode tracking control of nonlinear uncertain mass–spring system with experimental verifications. *Int J Adv Robot Syst* 2019; 16: 1729881419828176.
12. Wang B, Shi P and Karimi HR. Fuzzy sliding mode control design for a class of disturbed systems. *J Franklin Inst* 2014; 351: 3593–3609.
13. Furat M and Eker. Experimental evaluation of sliding-mode control techniques. *Ç Üniv Müh Mim Fak Derg* 2012; 27: 23–37.
14. Bucak Voloşencu C, Küçük S, et al. An in-depth analysis of sliding mode control and its application to robotics. In: Voloşencu C, Küçük S, Guerrero J, et al. (eds) *Automation and control*. London: IntechOpen, 2020, pp. 4–22.
15. Lan J, Zhao X and Sun C. Data-driven sliding mode control for partially unknown nonlinear systems. *arXiv preprint arXiv:240316136*. 2024.
16. Mandalapu A, Coleman J and Crassidis A. Model-free sliding mode control for coupled square and non-square MIMO systems. In: *Proceedings of the 11th International Conference of Control Systems, and Robotics (CDSR 2024)*, Chestnut Conference Centre - University of Toronto, Toronto, Canada, June 2024, pp.1–8. DOI: 10.11159/cdsr24.120.
17. Chaudhary A and Bhushan B. Design of a sliding mode controller (SMC) based on reaching law. *Int J Comput Appl* 2018; 182: 45–49.
18. Zhu Q. Complete model-free sliding mode control (CMFSMC). *Sci Rep* 2021; 11: 22565.
19. Li H, Dou L, Su Z, et al. A new modeling reference direct adaptive sliding mode control for electromechanical actuator. In: *Proceedings of the 48th IEEE conference on decision and control (CDC) held jointly with 2009 28th Chinese Control Conference*. Shanghai, China: IEEE, 2009, pp.6078–6082.
20. Kumar TD and Mija S. Dynamic SMC control scheme with adaptively tuned PID controller for speed control of DC motor. In: *2015 IEEE international conference on industrial technology (ICIT)*. Seville, Spain: IEEE, 2015, pp.187–191.
21. Kumar KV, Kumar TA and Ganesh V. Chattering free sliding mode controller for load frequency control of multi area power system in deregulated environment. In: *2016 IEEE 7th power India international conference (PII-CON)*. Bikaner, India: IEEE, 2016, pp.1–6.
22. Gamboa S, Leica P and Yumiguano E. Speed-droop governor based on sliding mode control (SMC) for generators in interconnected power system. In: *2023 IEEE seventh ecuador technical chapters meeting (ECTM)*. Ambato, Ecuador: IEEE, 2023, pp.1–6.
23. Doğruer T, Yüce A and Tan N. PID controller design based on Reference Model in fractional order control systems. *Bilge Int J Sci Technol Res* 2017; 1: 52–58.
24. Wang Q, Wang X and Cai Z. Model reference based on sliding mode control. In: *2009 IEEE international conference on intelligent computing and intelligent systems*. Shanghai/China: IEEE, 2009, pp.738–741.
25. Nguyen LP and Tran TT. A family of sliding mode controllers: analysis, design and comparison. In: *2023 International conference on system science and engineering (ICSSE)*. Ho Chi Minh, Vietnam: IEEE, 2023, pp.462–467.
26. Kamal S, Chalanga A and Bandyopadhyay B. Multivariable continuous integral sliding mode control. In: *2015 International workshop on recent advances in sliding modes (RASM)*. Istanbul, Turkey IEEE, 2015, pp.1–5.
27. Puentes K, Morales L, Pozo-Espin DF, et al. Enhancing control systems with neural network-based intelligent controllers. *Emerg Sci J* 2024; 8: 1243–1261.
28. Baghdasaryan M and Hovhannisyan V. Stability assessment of an ore mill electric drive using machine learning. *HighTech Innov J Optim Theory Appl* 2024; 5: 213–230.
29. Abidova E. Visualization and analysis method of defect manifestation in electromechanical equipment. *Emerg Sci J* 2024; 8: 1293–1303.
30. Hou H, Yu X, Xu L, et al. Finite-time continuous terminal sliding mode control of servo motor systems. *IEEE Trans Ind Electron* 2020; 67: 5647–5656.
31. Rsetam K, Cao Z, Man Z, et al. GPIO-Based continuous sliding mode control for networked control systems under communication delays with experiments on Servo Motors. *J Autom Sin* 2025; 12: 99–113.
32. Khan RFA, Rsetam K, Cao Z, et al. Singular perturbation-based adaptive integral sliding mode control for flexible joint robots. *IEEE Trans Ind Electron* 2023; 70(10): 10516–10525.
33. Zheng Y, Cao Z, Rsetam K, et al. Extreme learning machine-based super-twisting integral terminal sliding

- mode speed control of permanent magnet synchronous motors. *IET Control Theory Appl* 2024; 18: 2524–2539.
34. Rsetam K, Cao Z and Man Z. Design of robust terminal sliding mode control for underactuated flexible joint robot. *IEEE Trans Syst Man Cybern Syst* 2022; 52: 4272–4285.
35. Rsetam K, Cao Z and Man Z. Cascaded-extended-state-observer-based sliding-mode control for underactuated flexible joint robot. *IEEE Trans Ind Electron* 2020; 67(12): 10822–10832.
36. Ma'arif A and Çakan A. Simulation and arduino hardware implementation of dc motor control using sliding mode controller. *J Robot Control* 2021; 2: 582–587.
37. Ogata K (ed.). *Modern control engineering*. 3rd ed. Robins, 1997. pp.1013.